SF2956 Project

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Abstract

Wearable devices such as inertial measurement units (IMUs) are increasingly being used to monitor human motion in everyday environments. This project explores whether data collected from an IMU can be used to accurately distinguish different human activities such as walking, sitting, and standing. By representing the multi-sensor time-series data through topological methods, we investigate how the underlying geometry of motion patterns can be captured more effectively. The use of Topological Data Analysis (TDA) leads to improved classification accuracy compared to applying an SVM directly to the original dataset. These results suggest that topology-based representations can uncover structural information hidden in sensor data and hold promise for developing more robust activity recognition systems in the future.

The reproducible code and complete results are publicly available at `https://github.com/12ccc/SF2956-project`.

1 Theoretical Background

In this section some theoretical properties needed for our analysis will be introduced. All definitions in this part are taken from the lecture notes by Chacholski [1].

1.1 Distance

We start by defining a notion of distance between elements or data points, which serves to generalize the idea of how data points are related.

Definition: Let X be a set. A *distance* on X is a function

$$d : X \times X \to [0, \infty)$$

satisfying the following conditions for all $x, y \in X$:

(1) Symmetry: $d(x, y) = d(y, x)$;

(2) Reflexivity: $d(x, x) = 0$.

A distance d is called a *pseudometric* if, in addition, it satisfies:

(3) Triangle inequality: $d(x,y) + d(y,z) \geq d(x,z), \quad \forall x,y,z \in X.$

A pseudometric is called a *metric* if it also satisfies the following property:

(4) Identity of indiscernibles: $d(x,y) = 0.$

That is, $d(x,y) = 0$ holds if and only if $x = y$.

A set equipped with a chosen distance function is called a distance space. We denote by:

$$D(X), \quad PM(X), \quad M(X)$$

the sets of all distances, pseudometrics, and metrics on X, respectively. These sets satisfy the following inclusions:

$$M(X) \subseteq PM(X) \subseteq D(X).$$

In the following, we will describe the following distances. The Hamming, Cityblock (ℓ_1), Euclidean (ℓ_2), the general ℓ_p with $p \geq 1$, and Chebyshev (ℓ_∞) distances are genuine metrics, i.e., they belong to $M(X)$. The cosine and correlation distances do not satisfy the triangle inequality (and hence are not pseudometrics), so they belong only to $D(X)$.

Cityblock metric (ℓ_1):

$$d_1(x,y) = \|x-y\|_1 = \sum_{i=1}^{n} |x_i - y_i|. \tag{1}$$

Euclidean metric (ℓ_2):

$$d_2(x,y) = \|x-y\|_2 = \sqrt{\sum_{i=1}^{n} |x_i - y_i|^2}. \tag{2}$$

Hamming metric (ℓ_0):

$$d_0(x,y) = \|x-y\|_0 = \sum_{i=1}^{n} \mathbf{1}_{\{x_i \neq y_i\}}, \tag{3}$$

General ℓ_p metric ($p > 0$):

$$d_p(x,y) = \|x-y\|_p = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}. \tag{4}$$

Chebyshev metric:

$$d_\infty(x,y) = \|x-y\|_\infty = \max\{|x_1 - y_1|, \ldots, |x_n - y_n|\}. \tag{5}$$

Cosine distance:

$$d(x,y) = \begin{cases} 1 - \dfrac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2}\,\sqrt{\sum_{i=1}^{n} y_i^2}}, & \text{if } x \neq 0 \text{ and } y \neq 0, \\ \infty, & \text{otherwise.} \end{cases} \tag{6}$$

Correlation distance:

$$d(x,y) = 1 - \frac{(x-\bar{x})\cdot(y-\bar{y})}{\|x-\bar{x}\|_2\,\|y-\bar{y}\|_2}, \tag{7}$$

where

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

denote the means of x and y, respectively, and

$$x_c = x - \bar{x}, \quad y_c = y - \bar{y}$$

are the centered versions of x and y.

In this project, we only consider and compare the Euclidean, Cityblock, Chebyshev distances and the cosine distance.

1.2 Extensions

We now consider the notion of distance between subsets of a set X, rather than only between individual elements.

Let X be a set. The symbol $\Delta[X]$ denotes the set of all non-empty finite subsets of X. We identify an element $x \in X$ with the corresponding singleton $\{x\} \in \Delta[X]$. Using this identification, we consider X as a subset of $\Delta[X]$. In this section we discuss various ways of extending arbitrary distances on X to distances on $\Delta[X]$.

By restricting a distance on $\Delta[X]$ along the inclusion $X \hookrightarrow \Delta[X]$, we obtain a distance on X. We think about this restriction procedure as a function

$$\phi : D(\Delta[X]) \to D(X).$$

Any section to this restriction is called an *extension along* $X \hookrightarrow \Delta[X]$. Thus an extension along $X \hookrightarrow [X]$ is a function$\phi : D(X) \to D(\Delta[X])$, extending a distance d on X to a distance ϕd on $\Delta[X]$, which makes the following diagram commutative:

$$\begin{tikzcd} D(X) \arrow[r, "\phi"] \arrow[dr, "id"'] & D(\Delta[X]) \arrow[d, "\text{rest}"] \\ & D(X) \end{tikzcd}$$

Explicitly, $\phi : D(X) \to D(\Delta[X])$ is an extension, if for every distance d on X and all x and y in X:

$$d(x, y) = \phi d(x, y).$$

If d is a distance on X, then for non-empty finite subsets $A, B \subseteq X$, the following are distances on $\Delta[X]$:

- **Single-linkage extension of d:**

$$sd(A, B) := \min\{d(a, b) \mid a \in A, b \in B\}. \tag{8}$$

- **Complete-linkage extension of d:**

$$cd(A, B) := \begin{cases} \max\{d(a, b) \mid a \in A, b \in B\}, & \text{if } A \neq B, \\ 0, & \text{if } A = B. \end{cases} \tag{9}$$

- **Average-linkage extension of d:**

$$ad(A, B) := \begin{cases} \dfrac{\sum_{a \in A, b \in B} d(a, b)}{|A||B|}, & \text{if } A \neq B, \\ 0, & \text{if } A = B. \end{cases} \tag{10}$$

- **Ward linkage extension of d:** Assume d is a restriction along $f : X \to \mathbb{R}^n$ of a metric l, then

$$w_d(A, B) := \frac{|A||B|}{|A| + |B|} l(m_A, m_B), \tag{11}$$

  where

$$m_A = \frac{1}{|A|} \sum_{a \in A} f(a), \quad m_B = \frac{1}{|B|} \sum_{b \in B} f(b).$$

The distances s_d, c_d, a_d, and w_d are all valid distances on $\Delta[X]$ and are commonly used in the construction of dendrograms.

1.3 Dendrograms

A *dendrogram* is a tame set D parametrised by $[0, \infty)$, i.e., a sequence of sets D_t indexed by $t \in [0, \infty)$ together with a sequence of functions

$$D_{s<t} : D_s \to D_t,$$

called *transitions*, indexed by pairs $s < t$.

These transitions satisfy the following properties:

1. For any $r < s < t$, the following diagram commutes:

$$\begin{tikzcd} D_r \arrow[r, "D_{r<s}"] \arrow[dr, "D_{r<t}"'] & D_s \arrow[d, "D_{s<t}"] \\ & D_t \end{tikzcd}$$

2. There exists a sequence of real numbers

$$0 = a_0 < a_1 < \cdots < a_n$$

such that $D_{s<t} : D_s \to D_t$ is an isomorphism whenever s and t lie in the same interval $[a_0, a_1), \ldots, [a_{n-1}, a_n), [a_n, \infty)$. We say that the sequence $0 = a_0 < a_1 < \cdots < a_n$ *discretises* the dendrogram.

3. The transition function $D_{s<t} : D_s \to D_t$ is surjective for all $s < t$ in $[0, \infty)$.

If only properties (1) and (2) are satisfied, then D is called a *tame set parametrised by* $[0, \infty)$.

1.4 Hierarchical clustering

In the previous section, we defined a dendrogram as a tame set parametrised by $[0, \infty)$, representing how clusters evolve over a continuous range of the scale parameter. We now describe how such a dendrogram can be constructed from a distance function. By induction (for details, [1]), an extension $\psi : D(X) \to D(\Delta[X])$ along $X \subset \Delta[X]$ induces a function that maps a distance d on X to a dendrogram ψd on X. This function is called **hierarchical clustering** with respect to ψ, and it is denoted by the same symbol:

$$\psi : D(X) \to \mathrm{Dend}(X).$$

Intuitively, the hierarchical clustering process transforms a distance function into a nested family of clusters, capturing the multi-scale structure of the dataset.

However, dendrograms have certain limitations. In particular, their computational complexity is at least $O(n^2)$, making them inefficient for large datasets. Moreover, dendrograms are not well suited for direct comparison across different data sources.

1.5 Simplicial Complex

A **simplicial complex** on U is a collection $\mathcal{K}$ of finite non-empty subsets of U that satisfies the following requirement: If U is in $\mathcal{K}$, then any non-empty subset of U is also in $\mathcal{K}$. Let $X \subseteq U$ be a subset. Two special types of simplicial complexes associated with X are as follows:

- **Discrete simplicial complex.** The collection

$$\mathcal{X} = \{\{x\} \mid x \in X\},$$

consisting of all singletons in X, is a simplicial complex denoted also by the symbol X and called the *discrete simplicial complex* on X.

- **Simplex on X.** The collection

$$[X] = \{\sigma \subseteq X \mid \sigma \text{ is finite and } \sigma \neq \emptyset\},$$

is also a simplicial complex, denoted by $[X]$ and called the *simplex on X*.

One simplicial complex used in this project is the **Vietoris–Rips complex**.

For a given distance d on X and a parameter $t \in [0, \infty)$, the **Vietoris-Rips complex** of (X, d) at scale t is defined as

$$\mathrm{VR}_t(d) = \left\{\sigma \subseteq X \,\middle|\, d(x_i, x_j) \leq t \text{ for all } x_i, x_j \in \sigma\right\}.$$

That is, $\mathrm{VR}_t(d)$ consists of all finite non-empty subsets of X whose pairwise distances are at most t. The family $\{\mathrm{VR}_t(d)\}_{t \geq 0}$ forms a filtration of simplicial complexes, encoding the full metric information of (X, d). Unlike partitions induced by ultra pseudometrics, Vietoris–Rips complexes retain the full information of d and enable topological analysis of data through homology.

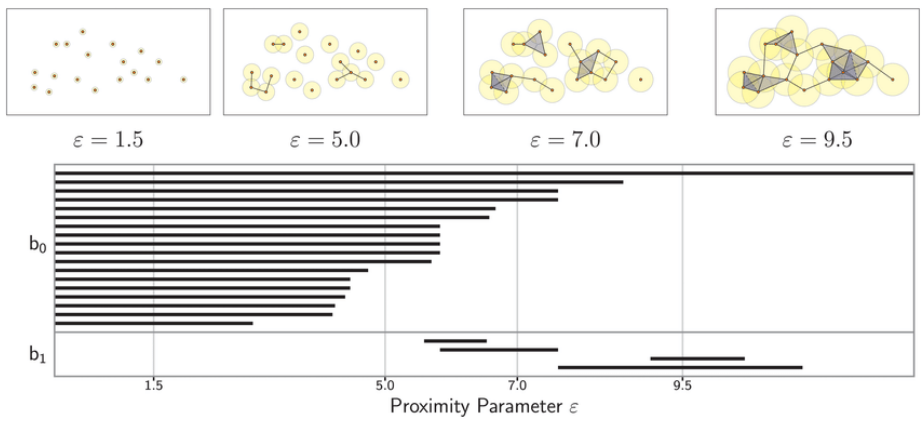


Figure 1: Example of Vietoris-Rips complex [3]

Here is an example of the Vietoris–Rips complex construction of 18 points. From left to right, we can observe that for small radius ϵ, there are only vertices and no edges; As the radius ϵ increases,edges begin to appear between nearby points, forming two connected components. These edges together with vertices, build higher-dimensional features that persist as ϵ grows. When the radius increases further, a loop (1-dimensional homology) emerges.

1.6 Homology

Let X be a simplicial complex and F a field. The *n-th homology group* of X over F is defined as the quotient

$$H_n(X, F) := \ker(\delta_n) / \operatorname{im}(\delta_{n+1}),$$

where $\delta_n : FX_n \to FX_{n-1}$ is the *boundary operator*. Here, X_n denotes the set of n-dimensional simplices of X, and the vertices of X are ordered as $X_0 = \{v_0 < v_1 < \cdots < v_k\}$.

The boundary operator δ_n is defined by

$$\delta_n = \sum_{i=0}^{n} (-1)^i \, F d_i,$$

where each $d_i : X_n \to X_{n-1}$ removes the i-th vertex:

$$d_i(x_0 < x_1 < \cdots < x_n) = (x_0 < \cdots < \widehat{x_i} < \cdots < x_n).$$

Elements of $\ker(\delta_n)$ are called *cycles* when the dimension is clear. Elements of $\mathrm{im}(\delta_{n+1})$ are called *boundaries*. Intuitively, homology measures how many more cycles than boundaries there are, i.e., it captures the independent "holes" that cannot be filled by higher-dimensional simplices.

In this project, homology is used to quantify the number of holes in the Vietoris–Rips complexes at different scales. Concretely, we focus on the following cases:

- H_0: Describes the number of connected components.
- H_1: Describes the number of one-dimensional holes, typically loops or cycles.

Homology thus provides a bridge from the combinatorial structure of simplicial complexes to the topological features of the underlying data space, transforming it into a vector space representation.

1.7 Barcodes

Barcodes provide a compact and visual representation of the homological information extracted from data. Given a simplicial complex X and a field F, the n-th homology group $H_n(X, F)$ captures the n-dimensional topological features of X. As the radius or scale increases, these features appear and disappear, giving rise to the *birth* and *death* times of homological structures.

Each homological feature thus has an associated interval $[a_i, b_i)$, where a_i denotes its birth time and b_i its death time.

Formally, we define:

$$[a_i, b_i)(t) = \begin{cases} F, & \text{if } t \in [a_i, b_i), \\ 0, & \text{otherwise.} \end{cases}$$

Each bar represents a vector space that is active during its corresponding interval and vanishes elsewhere.

For every $s < t$ in $[0, \infty)$, the transition map

$$[a_i, b_i)_{s<t} : [a_i, b_i)(s) \longrightarrow [a, b)(t)$$

is defined by

$$[a_i, b_i)_{s<t} = \begin{cases} \mathrm{id}_F, & [a_i, b_i)(s) = [a_i, b_i)(t) = F, \\ 0, & \text{otherwise.} \end{cases}$$

More generally, if V is a *tame* vector space parameterized by $[0, 1)$ and is isomorphic to a finite direct sum of interval modules,

$$V \cong \bigoplus_{i=1}^{n} [a_i, b_i),$$

then the collection of pairs $\{(a_i, b_i) \mid i = 1, \ldots, n\}$ is called the **barcode** of V. Intuitively, the barcode encodes the lifespan of each homological feature in the filtration.

Figure 1 in section 1.5 illustrates the concept for the zeroth and first homology groups, as well as the connection between simplicial complexes, their corresponding barcodes, and persistence diagrams. The bars in b_0 represent the 0-dimensional homologies (connected components), while those in b_1 correspond to 1-dimensional homologies (loops or cycles).

The vertical lines in the barcode correspond to four different levels of the proximity parameter ε. The number of horizontal bars intersecting each vertical line gives the Betti numbers $b(\varepsilon) = (b_0(\varepsilon),\, b_1(\varepsilon))$.

1.8 Stable Rank

It is known that a dendrogram can be transformed into a tame parametrized vector space. Let V be a tame parametrized vector space, i.e., a family $\{V_t\}_{t\in[0,1)}$ of vector spaces whose structure changes only at finitely many values of t. From this vector space, one can compute the *stable rank*, which results in a piecewise-constant function (PCF). Compared to the dendrogram, which represents hierarchical relationships in a tree-like structure, the stable rank provides a compact, real-valued functional representation that is easier to analyze and compare numerically.

The *stable rank* of V is defined as:

$$\widehat{\mathrm{rank}}V(t) := \min\{\mathrm{rank}(W) \mid d_{\bowtie}(V, W) \leq t\}.$$

where $d_{\bowtie}(V, W)$ denotes the interleaving distance between the parametrized vector spaces V and W. At $t = 0$, the stable rank reduces to the usual rank: $\widehat{\mathrm{rank}}V(0) = \mathrm{rank}(V)$.

Intuitively, the stable rank $\widehat{\mathrm{rank}}V(t)$ measures the minimal number of persistent features of V that remain after smoothing it with tolerance t. It provides a robust summary that quantifies how the homological features of the data persist under noise.

Moreover, there exists a stability relation between the interleaving distances of parametrized vector spaces and their corresponding stable rank functions:

$$d_{\bowtie}(V, W) \geq d_{\bowtie}(\widehat{\mathrm{rank}V}, \widehat{\mathrm{rank}W}).$$

When computing the stable rank, we move from a tame parametrized vector space to a piecewise-constant function. If V is discretized by $0 = a_0 < a_1 < \cdots < a_n < 1$, then:

$$\dim(V_t) : [0, 1) \to \mathbb{N}, \quad t \mapsto \dim(V_t),$$

and for each interval $[a, b)$ corresponding to a bar in the barcode, we have:

$$\dim_{[a,b)}(t) = \begin{cases} 1, & \text{if } t \in [a, b), \\ 0, & \text{otherwise.} \end{cases}$$

which records when the bar is alive in the filtration.

Here, It is important to clarify that **the stable rank is not the same as the dimension function**.

The stable rank depends only on the length of the bar, not on its birth time. For a bar of length $\ell = b - a$, its contribution to the stable rank is

$$\widehat{\text{rank}}_{[a,b)}(t) = \begin{cases} 1, & \text{if } t \in [0, \ell), \\ 0, & \text{otherwise.} \end{cases}$$

Thus, a bar contributes 1 to $\widehat{\text{rank}}(t)$ for all t smaller than its lifetime and 0 afterwards.

Thus, unlike the dimension function, which measures the number of bars alive at a given filtration parameter, the stable rank at scale t counts the number of bars with persistence (length) greater than t.

2 Dataset Description and Analysis

2.1 Dataset Description

The data used in this study are taken from the IMU-based Human Activity Recognition Dataset (version 3) by Tahir, Yara and Hamarash, Ibrahim Ismael (2025).

The dataset contains inertial measurements recorded using an IMU connected to a Raspberry Pi. During each recording session, the IMU captured six sensor channels: three-axis linear acceleration and three-axis angular velocity (gyroscope).

The recorded activities include walking upstairs, walking downstairs, walking, jogging, sitting, and standing. The activity label for each sample was originally stored as text and later converted to numerical labels (1–6) for ease of analysis.

A detailed description of the dataset variables is provided in following table:

Field	Description
`ax`, `ay`, `az`	Accelerometer readings along x,y,z
`gx`, `gy`, `gz`	Gyroscope readings along x,y,z
`activity_id`	Activity numeric code (1-6)

Table 1: fields in each IMU recording

2.2 Data Visualization

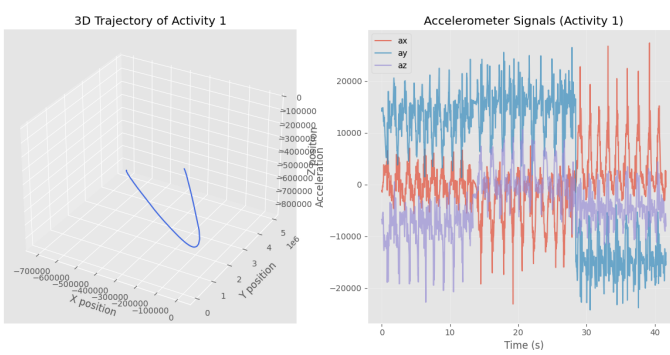
Given the triaxial accelerometer signals $\mathbf{a}(t) = [a_x(t), a_y(t), a_z(t)]$, the subject's motion trajectory can be approximately reconstructed using a simple numerical double integration. the velocity and position at each time step t_k are estimated as

$$\mathbf{v}(t_k) = \sum_{i=1}^{k} \mathbf{a}(t_i)\,\Delta t, \qquad \mathbf{p}(t_k) = \sum_{i=1}^{k} \mathbf{v}(t_i)\,\Delta t, \tag{12}$$

where $\mathbf{v}(t_k)$ and $\mathbf{p}(t_k)$ denote the velocity and position, respectively. Prior to integration, the acceleration signals are mean-centered to remove static biases such as sensor drift or gravity offset.

Although this approach does not yield metrically precise positions, it provides an *approximate spatial trajectory* that captures the dynamic patterns of different activities.

By visualizing both the 3D trajectories $\mathbf{p}(t)$ and the accelerometer time series $\mathbf{a}(t)$ for each activity, distinct motion signatures can be observed:



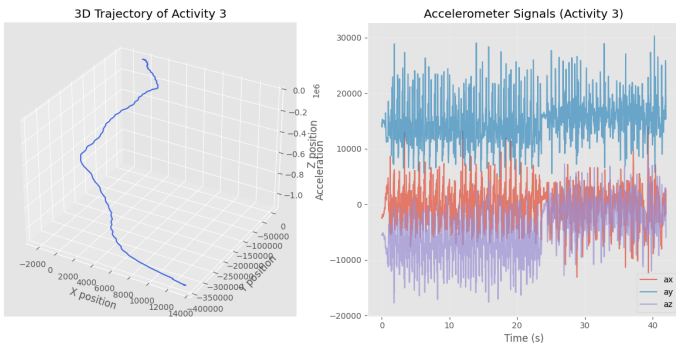


Figure 2: visualization examples

3 Topological Data Analysis Pipeline

We first split the dataset into training and testing subsets using an 8:2 ratio.

Point Cloud Construction. The time-series data were then segmented into overlapping sliding windows to form point clouds, enabling topological data analysis (TDA) to capture both the shape and temporal structure of motion trajectories.

- **Window size:** 200 samples
- **Step size:** 50 samples (high overlap for dense sampling)

Only windows with consistent activity labels were retained. Each resulting point cloud has shape $(\text{window_size}, d)$, where d is the number of features (e.g., accelerometer axes). For further analysis, point clouds can be flattened into 1D vectors of length $\text{window_size} \times d$.

Distance Matrix For each distance metric (Euclidean, cityblock, Chebyshev, and cosine), we first compute the distance matrix between all point clouds. This matrix serves as the foundation for both clustering and topological analysis.

Linkage and Dendrogram Construction. Using multiple linkage methods (single, complete, average, and ward), we construct dendrograms to examine whether individuals from the same group cluster together before merging with other groups.

Vietoris-Rips Complex. Alternatively, we construct Vietoris-Rips complexes directly from the point clouds using the same distance metrics. This approach captures the topological structure of the data across different scales.

Persistent Homology. From the dendrograms or the Vietoris-Rips complexes, we compute persistent homology (H_0 and H_1) to extract topological features such as connected components and loops. These barcodes summarize the shape and connectivity of the trajectories.

Piecewise Constant Function (PCF). The persistent homology barcodes are converted into piecewise constant functions (PCFs), which provide a fixed-length numerical representation of the topological features.

Kernel Construction. Using the PCFs, we construct kernels to define similarity measures between point clouds. These kernels allow us to compare trajectories based on their topological structure. Precomputed kernel matrices were constructed as:

$$K_{\text{train}} = X_{\text{train}} X_{\text{train}}^T, \tag{13}$$

$$K_{\text{test}} = X_{\text{test}} X_{\text{train}}^T. \tag{14}$$

Classification. Finally, the kernels are used in classification tasks with SVM method,enabling the discrimination of different activities or subject groups based on their topological signatures.

4 Results and Analysis

4.1 Preliminary Classification Evaluation

Before performing topological data analysis (TDA), we evaluated several standard machine learning classifiers on the same dataset. The comparison results are summarized in Table 2. Among all tested models, Although some models such as MLP achieved slightly higher accuracy, the **SVM** and **SVM** provided good generalization capability and shorter training time, they were therefore selected for the following TDA-based classification experiments.

Table 2: Performance comparison of different classification models

Model	**Description / Key Parameters**	**Accuracy**
SVM	Support Vector Machine (RBF kernel)	0.8176
KNN	k-Nearest Neighbors ($k = 10$)	0.8598
Random Forest	200 trees, random state = 42	0.8462
Logistic Regression	Max iterations = 1000	0.3170
MLP	Hidden layers (256, 128), max iter = 100	0.8561

4.2 Distance

To better understand the pairwise dissimilarities between samples, we visualized the distance distributions of the 32nd sample under four different metrics: Euclidean, Cityblock, Cosine, and Chebyshev (Figure 3).

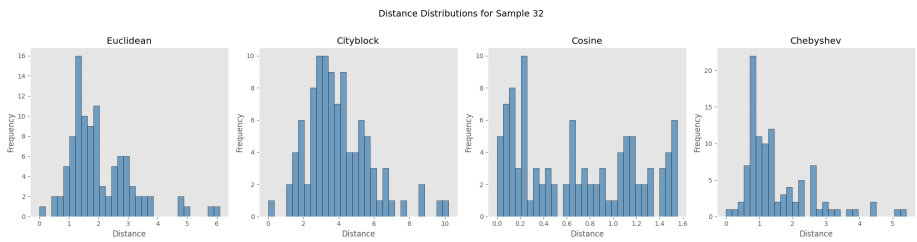


Figure 3: Distance distributions of the 32nd sample under different metrics.

Overall, the Euclidean, Cityblock, and Chebyshev metrics produce approximately unimodal and near-Gaussian distributions, indicating relatively uniform geometric separation among samples.

In contrast, the Cosine distance exhibits a multimodal pattern, suggesting the presence of several distinct directional groups in the feature space. This indicates that cosine distance is more sensitive to angular separation.

4.3 Persistence Diagram

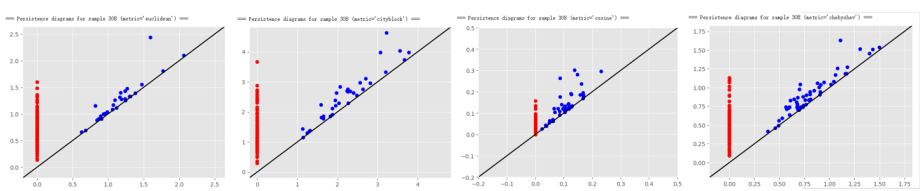


Figure 4: Persistence diagrams for sample 308

Figure 4 shows the persistence diagrams (H0 in red and H1 in blue) computed for the same sample using four distance metrics: Euclidean, cityblock, cosine, and Chebyshev.

Across all metrics, H0 features are born near zero and die at varying scales, indicating that the data initially forms many isolated points that gradually merge into coherent clusters as the filtration scale increases. This pattern reflects a well-structured point cloud rather than random noise.

In contrast, H1 features exhibit distinct persistence behaviors depending on the metric. For this sample, the Euclidean and cityblock metrics yield clearer separation of persistent topological features, several H1 points lie well above the

diagonal, suggesting the presence of stable one-dimensional cycles that persist over a wide range of scales.

4.4 Barcodes

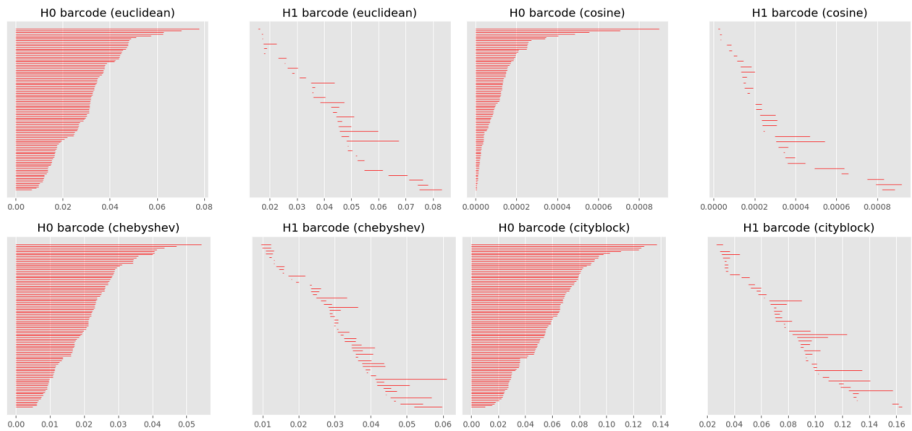


Figure 5: Barcodes for sample 614

Figure 5 illustrats the persistence of topological features in dimensions H_0 and H_1 under different distance metrics. In the H_0 barcode, a few longer bars persist across wider scales, indicating the presence of stable and well-separated clusters within the underlying point cloud. The H_1 barcode shows several bars with moderate persistence, suggesting that the data exhibits nontrivial one-dimensional cycles.

It is also observed that the cosine metric produces generally shorter barcodes, as it emphasizes angular similarity and compresses magnitude differences between samples.

4.5 Stable Rank

4.5.1 H0

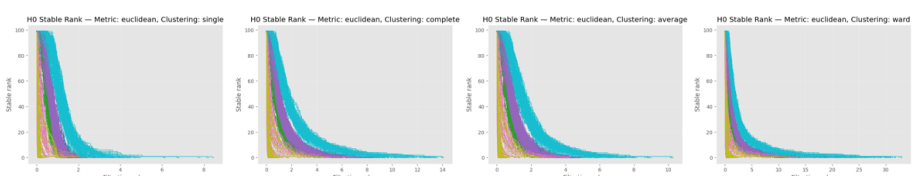


Figure 6: Stable Rank H0 with euclidean metric

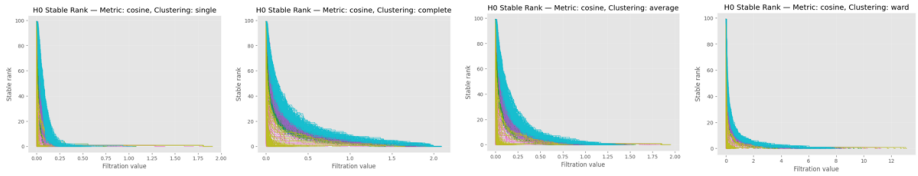


Figure 7: Stable Rank H0 with cosine metric

Figure 6, 7 illustrates the stable rank functions computed for different distance metrics, clustering methods with homology degrees H_0. the stable rank decreases as the filtration value increases, reflecting the merging of connected components.

Each curve represents one sample, color-coded by activity label. The distinct curve patterns across activity classes suggest that the stable rank captures meaningful topological differences in the underlying motion trajectories.

For H0 with Euclidean metrics, the single, complete and average metrics show similar results, while for the ward metric the curve moves quite a bit.

Comparing the Euclidean–complete and cosine–single combinations, The distinct patterns suggests that that the former captures more meaningful topological features, which will be reflected in the subsequent SVM results.

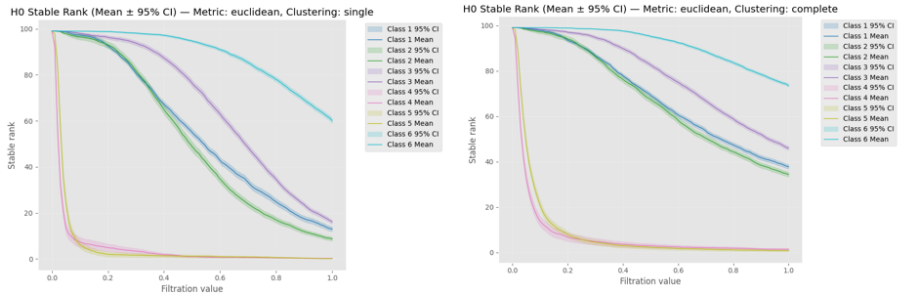


Figure 8: Stable rank averages and standard deviation

The figure shows the mean stable rank functions (with 95% CI) for each class.

4.5.2 H1

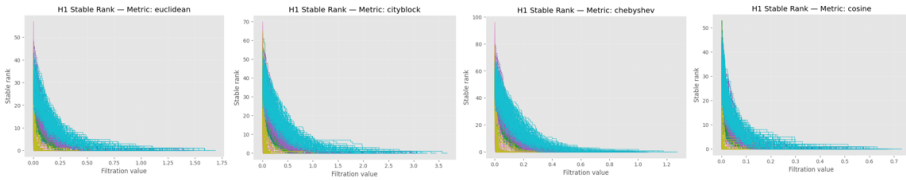


Figure 9: Stable Rank H1

Figure 9 shows the stable rank functions of H_1 computed using different distance metrics. The four metrics lead to similar results.

The curves are less distinct compared to H_0, which may indicate poorer class separability and thus a worse classification performance.

4.6 Classification

Here, only the (H0) features were used for classification, with SVM as the classifier.

Table 3: SVM&KNN Evaluation Summary using multiple linkage methods

Distance Metric	Clustering Method	KNN Acc	SVM Acc
cityblock	ward	**0.8423**	0.8115
chebyshev	ward	0.8423	0.8385
euclidean	ward	0.8308	0.8231
euclidean	complete	0.8308	0.8808
euclidean	average	0.8269	0.7923
cityblock	complete	0.8154	0.8038
chebyshev	complete	0.8154	**0.9231**
cityblock	average	0.8115	0.7346
chebyshev	average	0.8115	0.8346
chebyshev	single	0.8038	0.8500
euclidean	single	0.7885	0.8462
cityblock	single	0.7769	0.7885
cosine	average	0.6500	0.7154
cosine	ward	0.6462	0.7154
cosine	complete	0.6231	0.7038
cosine	single	0.5769	0.6038

Table 3 presents the SVM and KNN evaluation results across different distance metrics and clustering methods.

Overall, the cosine distance yielded relatively lower performance, suggesting that it may be less effective in separating classes in this feature space than other distance metric.

For the KNN classifier, the cityblock distance metric combined with the Ward clustering method achieved the highest accuracy of 0.8423, which is slightly lower than the original accuracy of 0.8598 reported in Table 2.

For the SVM classifier, the Chebyshev distance metric combined with the Complete clustering method achieved the highest accuracy of 0.9231, which is substantially higher than the original accuracy of 0.8176.

In conclusion, TDA provide a meaningful representation of the dataset, enabling classifiers to achieve high performance for this project.

Table 4: SVM classification performance using Vietoris–Rips (VR) method

Metric	**Accuracy**
Euclidean	0.8462
Cityblock	0.7885
Correlation	0.6500
Cosine	0.6038

The results indicate that in this project the VR-based features yield relatively lower classification accuracy (0.8462) than multiple linkage methods, but it's still better than the accuracy in table 2.

References

1. Wojciech Chacholski. *Topological Data Analysis*, Lecture Notes.

2. Tahir, Yara and Hamarash, Ibrahim Ismael. *IMU-based Human Activity Recognition Dataset 2025 (version 3)*. DOI: `10.17632/fcnmmsn857.3`.

3. `https://www.researchgate.net/publication/269876798_Topological_Data_Analysis_of_Biological_Aggregation_Models`